

Jonathan Fang

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Education

Georgia Institute of Technology

M.S. Computer Science

Atlanta, GA

Aug. 2026 – May 2028

University of Wisconsin-Madison

B.S. Electrical and Computer Engineering

Madison, WI

May 2025

- Relevant Classes: Microprocessor Systems, Digital System Design and Synthesis, Semiconductor Analysis, Artificial Intelligence, Computer Graphics, Compilers, Testable Design of Digital Systems

Technical Skills

Languages: C/C++, Python, SystemVerilog, Java, Bash, JavaScript

Tools: Linux (Ubuntu), Cursor, Git, GDB, Jira, Quartus Prime, ModelSim, LTSpice, Altium, Oscilloscope, JTAG

Experience

Test Development Engineer

Cisco · Product Operations

March 2026 – Present

San Jose, CA

- Modernize manufacturing diagnostic tests for Catalyst 9400 switches across 3 board types into Cisco's current automated test framework, keeping factory test coverage consistent as boards move through production
- Validate and debug test sequences directly on hardware at production stations, partnering with the diagnostics team and CM engineers to resolve failures and keep manufacturing lines moving
- Connect product traceability data (serial numbers, assembly genealogy, cloud IDs) into the test flow to keep unit records aligned across the supply chain

AI Engineering Research Fellow

Handshake AI

Dec. 2025 – Present

Contract/Part-time

- Author and verify self-contained, executable analog circuit-design and SPICE simulation tasks, spanning active filters, multi-stage amplifiers, and behavioral protection and control-logic systems across DC, AC, transient, and Fourier analyses

Academic Tutor

AJ Tutoring

Sept. 2025 – Jan. 2026

Menlo Park, CA

- Managed caseload of 15 students across Algebra 2 Honors through Calculus, designing personalized curriculum and learning plans tailored to individual learning styles and academic goals

IT Technical Intern

University of Wisconsin - Madison Department of Information Technology

Oct. 2023 – Dec. 2024

Madison, WI

- Managed authentication infrastructure and enterprise software for 40,000+ users, maintaining 99.9% system uptime

Embedded Research Lead

UW Transcend

Sept. 2021 – May 2022

Madison, WI

- Led 4-person team developing real-time FPGA-based image processing system, coordinating sprints and managing development pipeline using Jira and Agile methodology
- Optimized critical path timing to increase clock frequency from 125MHz to 150MHz (20% performance gain) while reducing FPGA resource utilization from 85% to 68%, validated through hardware testing protocols

Projects

PSoC6 Embedded Multiplayer Console | C, FreeRTOS, SPI, I2C, UART

- Designed modular game console platform with hot-swappable cartridges using FreeRTOS firmware architecture, enabling rapid game deployment across multiple titles
- Developed low-level device drivers for 16-bit LCD over SPI/I2C, integrating GPIO interrupts for precise system state management and real-time input handling

Partial Scan Test Design Optimization | SystemVerilog, Synopsys, TetraMax, JTAG

- Designed and optimized partial scan architecture for b14 benchmark circuit, achieving 46% improvement in test efficiency over full scan implementation while reducing chain length by 80%
- Reduced chip area by 7% through strategic circuit optimization while maintaining 41%+ fault detection coverage